



HullProd

Geometry-Based Producibility Screening
for Ship Hull Surfaces

Theory, Implementation and User Manual

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This manual describes the executable behavior of the public v1.0.1 release. Mathematical definitions are included only where they can be traced to the versioned documentation or implementation. The tagged repository source remains authoritative for executable behavior.

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How to read this manual

This manual documents the implemented assessment pipeline of HullProd v1.0.1: geometry conventions, mathematical definitions, native-BRep and triangle-mesh realizations, command-line and Python usage, output files, and verification examples.

The companion paper by Serani and Maki [1] is the single scientific reference for the formulation and benchmark study. Source-code paths, release identifiers, figure provenance and archived execution records are documented separately as reproducibility information, not as additional scholarly references. The tagged repository remains authoritative for executable behavior.

Three labels distinguish the outputs already present in this release: **Canonical** for the recommended scientific contract, **Auxiliary** for supporting or representation-diagnostic quantities, and **Experimental** for opt-in compatibility paths excluded from the recommended signature. Metric-specific validity is a separate numerical qualification.

The illustrations retain the original paper artwork. The comparison chart uses unchanged archived release data. The sphere tutorial uses a geometry bundled with the public package and provides a complete procedure for creating new output files. Its reference values are analytical targets; no new HullProd execution is claimed for this manual edition.

Important limitation

A finite number is not automatically a canonical result. HullProd reports metric-specific validity and representation provenance. In particular, a triangle-mesh result can be finite and scientifically useful while still being explicitly labelled `mesh_representation_sensitive`. Always interpret the signature together with `validity.json` and `provenance.json`.

1. Purpose and supported capabilities

1.1. What HullProd is

HullProd is an open Python package for geometry-based producibility screening of ship hull surfaces. Given either a native IGES/STEP boundary representation (BRep) or a triangulated STL/OBJ/PLY surface, it computes a compact, dimensionless signature of double-curvature intensity, curvature sign, and areal composition. The same assessment also produces local surface fields, static plots, machine-readable validity, and representation provenance.

The practical question is intentionally narrower than full manufacturing simulation:

Given the represented external hull surface, how much non-developable curvature is present, what is its sign, where is it located, and how much of the valid represented area belongs to flat, singly curved, elliptic, or saddle/reverse classes?

The package provides early-stage geometric screening, reproducible benchmark comparison, and localized geometry diagnostics. It reports descriptors and their validity rather than inferring production data that are absent from the supplied surface.

1.2. Release snapshot

This manual is synchronized to the public release **v1.0.1**, dated 3 September 2026. The package is installable from PyPI as **hullprod** and supports CPython 3.10–3.12. Version 1.0.1 is a metadata-only patch over 1.0.0: it corrects the BEAM funding acknowledgement and introduces no scientific, numerical, API, or software-behavior change.

Canonical v1 behavior

The recommended v1 signature is fixed as

$$\mathbf{I}_{\text{sig}} = \left[I_D, I_D^+, I_D^-, a_C^{\text{flat}}, a_C^{\text{single}}, a_C^{\text{elliptic}}, a_C^{\text{saddle}} \right]^T.$$

The package does not collapse these seven components into an uncalibrated universal scalar producibility score.

1.3. Design principles

The release embodies the following design decisions:

- geometry-only quantities must not be presented as direct fabrication-cost or labor predictors;

- native CAD and triangulated surfaces are distinct numerical realizations and their provenance is never hidden;
- source CAD, when available, is evaluated directly rather than silently converted to a mesh for canonical global metrics;
- normalization must be dimensionally explicit and traceable through a declared or reproducible reference length;
- invalid, singular, or numerically unconverged regions must not be silently zero-filled into the global integral;
- validity is metric-specific: one unavailable quantity does not invalidate the complete assessment;
- recommended, auxiliary, diagnostic, and experimental quantities remain separate in both API and report semantics;
- local fields and global scalars are produced together so that a number can be spatially interpreted rather than treated as a black-box score.

1.4. Supported representations

Table 1.1.: Supported v1 geometry representations.

Representation	Formats	Numerical realization	Interpretation
Native BRep	.iges, .igs, .step, .stp	OpenCascade surface derivatives and trimmed-domain quadrature	Canonical realization when native CAD is available
Triangle mesh	.stl, .obj, .ply	Discrete curvature reconstruction on the supplied triangulation	Valid for the represented mesh; derivative-based quantities remain representation-sensitive

Unsupported formats fail explicitly. HullProd does not silently fit CAD to a mesh, does not silently tessellate a BRep for canonical integration, and does not guess the physical length unit of STL/OBJ/PLY files.

1.5. Capability matrix

Capability	Status	Current meaning
Native IGES/STEP assessment	Canonical	Direct BRep derivatives and trimmed-domain numerical integration.
STL/OBJ/PLY assessment	Canonical	Discrete mesh curvature on the supplied triangulated representation.
Recommended seven-component signature	Canonical	Total/signed double-curvature intensity plus four valid-area class fractions.

Capability	Status	Current meaning
Metric-specific validity	Canonical	Machine-readable status, reason, valid/represented area, backend and convergence metadata.
Representation provenance	Canonical	Input hash, version, backend, reference-length policy, thresholds, geometry quality, units, warnings and timings.
Distributed surface fields	Canonical	VTP and CSV exports for developability density, sign components, classes and available curvature fields.
Static HTML report	Canonical	Self-contained user summary with signature, plots, validity, provenance and output links.
Automatic reference length	Canonical	Principal-axis projected span with deterministic isotropic fallback; never identified as ship length between perpendiculars.
Thresholded developability area ratio	Auxiliary	Redundant with elliptic plus saddle class area under the common threshold.
Mesh-quality and local plate twist	Auxiliary	Representation diagnostics; not geometry-invariant signature components.
Curvature energy/fairness, section waviness	Experimental	Screened quantities available only by explicit opt-in and excluded from the recommended signature.

1.6. Scope of the reported results

HullProd does not predict fabrication cost, man-hours, shipyard productivity, plate-forming route, production schedule, or process feasibility. It does not perform panelization, seam placement, forming simulation, structural scantling assessment, geometry repair, or hull-form optimization. A low value of one metric is therefore not a universal statement that a vessel will be cheaper or easier to build in every yard.

The v1 result consists of geometric descriptors of the represented external surface, distributed fields, and explicit validity and provenance. This is the interpretation used throughout the manual and in the companion paper [1].

2. Geometry and mathematical conventions

2.1. Represented surface and valid domains

Let the represented external hull surface be $S \subset \mathbb{R}^3$. HullProd treats the BRep and mesh backends as two different numerical realizations of this surface. A native BRep is evaluated on trimmed parametric faces; a mesh is evaluated on its supplied vertices, triangles and connectivity.

Not every represented point must be valid for every differential quantity. For Gaussian-curvature metrics, denote by $S_K \subseteq S$ the valid represented domain and by

$$A_K = \int_{S_K} dA$$

its valid area. The developability intensities are normalized by A_K , not by an area that includes invalid points silently treated as zero. For curvature classification, let S_C be the valid classified domain and $A_C = \int_{S_C} dA$ its area. Class fractions use A_C . The two valid domains may coincide, but that coincidence should not be assumed when reading validity records [1].

Important limitation

The valid domain is part of the result. If an open mesh boundary makes Gaussian curvature invalid at boundary vertices, those vertices are excluded rather than assigned $K = 0$. Likewise, a native BRep integral that is non-integrable or numerically unconverged is not promoted to a canonical finite value.

2.2. Principal, mean and Gaussian curvature

At a regular point, let k_1 and k_2 denote the principal curvatures. HullProd uses the conventional mean and Gaussian curvatures

$$H = \frac{k_1 + k_2}{2}, \quad K = k_1 k_2.$$

The sign of H depends on the surface orientation convention; the sign of K is orientation invariant. The retained signature uses H only to distinguish quasi-flat from singly curved regions when K is near zero. The double-curvature intensity itself is based on $|K|$ and is therefore independent of the sign convention for the surface normal.

For interpretation:

- $K = 0$ throughout a sufficiently regular surface region is the local differential condition for developability; a single zero-curvature sample does not establish developability of an entire patch;

- $K > 0$ indicates elliptic/synclastic double curvature;
- $K < 0$ indicates saddle/reverse/anticlastic double curvature.

2.3. Reference length and nondimensionalization

Curvature is dimensional: H has dimensions L^{-1} and K has dimensions L^{-2} . HullProd therefore introduces a positive reference length L_{ref} and works with normalized densities such as $|K|L_{\text{ref}}^2$.

An explicit user value supplied with `--lref` is authoritative and is interpreted in the working length unit of the input geometry. For native CAD, the source/working unit can be reported from the BRep import. For STL/OBJ/PLY, no physical unit is declared by the file format and HullProd reports `input-length units`.

If no explicit length is supplied, HullProd computes a geometric convenience scale from controlled geometry samples. With centered samples $\mathbf{x}_i - \bar{\mathbf{x}}$, it forms the covariance matrix and takes the extent along the dominant eigenvector. If the leading eigenvalue is numerically non-unique, with relative eigengap not exceeding 10^{-5} , the deterministic fallback is

$$L_{\text{ref}} = 2 \max_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2.$$

This automatic value is recorded as `auto_principal_span`. It is rigid-motion invariant and scale covariant, but it is not inferred or reported as L_{pp} .

If an explicit L_{ref} differs from the automatic geometric span by a factor below 0.1 or above 10, HullProd issues a nonfatal plausibility warning. The user value remains unchanged; the software never guesses a unit conversion.

2.4. Quasi-zero thresholds

The curvature classes require numerical thresholds for distinguishing values that are treated as effectively zero at the selected reference scale. HullProd uses

$$\tau_H = \frac{h_f}{L_{\text{ref}}}, \quad \tau_K = \frac{k_f}{L_{\text{ref}}^2},$$

with default dimensionless factors

$$h_f = k_f = 10^{-4}.$$

These are numerical classification thresholds, not manufacturing tolerances, plate-forming limits, or acceptance criteria. Advanced users can override them, but any altered thresholds must be retained in provenance because they can change area-class composition.

2.5. Canonical and display representations

For native BRep input, global retained metrics are computed from direct CAD derivatives and trimmed-domain quadrature. A separate tessellation may be generated for visualization and field transport, but the global BRep integrals do not depend on that display tessellation.

For mesh input, the triangulation itself is the numerical representation used for differential reconstruction. Resolution, connectivity, aspect ratio, boundaries and local mesh quality can therefore affect the result. HullProd reports these conditions rather than hiding them.

3. Recommended v1 theory and metrics

3.1. Developability-deviation density

The retained formulation starts from Gaussian curvature because exact developable surfaces satisfy $K = 0$ at regular points. HullProd defines the local dimensionless double-curvature or developability-deviation density

$$q_D(\mathbf{x}) = |K(\mathbf{x})|L_{\text{ref}}^2.$$

The density is zero on an exactly developable region and increases with the magnitude of double curvature at the chosen reference scale. It should be interpreted as a geometry descriptor, not as a forming-energy or cost density.

3.2. Total double-curvature intensity

The global total intensity is the valid-area average

$$I_D = \frac{1}{A_K} \int_{S_K} |K|L_{\text{ref}}^2 dA. \quad (3.1)$$

The area normalization is important: I_D is an average over the valid represented domain, not an extensive quantity that automatically grows with hull surface area.

Canonical v1 behavior

I_D is the primary scalar measure of the magnitude of double curvature in the retained v1 signature. It answers “how much normalized non-developable curvature is present on average?” without discarding the sign or areal composition, which are stored separately.

3.3. Positive and negative contributions

To distinguish elliptic and saddle/reverse behavior, define

$$q_D^+(\mathbf{x}) = \max(K, 0)L_{\text{ref}}^2, \quad q_D^-(\mathbf{x}) = \max(-K, 0)L_{\text{ref}}^2,$$

and

$$I_D^\pm = \frac{1}{A_K} \int_{S_K} q_D^\pm dA. \quad (3.2)$$

Both quantities are nonnegative. By construction,

$$I_D = I_D^+ + I_D^-.$$

Thus I_D^+ measures elliptic/synclastic intensity, while I_D^- measures saddle/reverse/anticlastic intensity. The decomposition preserves physically meaningful sign information without allowing positive and negative Gaussian curvature to cancel in the total intensity.

3.4. Curvature-class area fractions

Using the quasi-zero thresholds τ_H and τ_K , HullProd partitions the valid represented area into four classes:

Table 3.1.: Curvature classes used in the v1 signature.

Component	Numerical condition	Geometric interpretation
a_C^{flat}	$ K \leq \tau_K$ and $ H \leq \tau_H$	both principal curvatures are quasi-zero
a_C^{single}	$ K \leq \tau_K$ and $ H > \tau_H$	one principal curvature is quasi-zero
a_C^{elliptic}	$K > \tau_K$	elliptic/synclastic double curvature
a_C^{saddle}	$K < -\tau_K$	saddle/reverse/anticlastic double curvature

For class c , with valid class subdomain $S_c \subseteq S_C$,

$$a_C^c = \frac{1}{A_C} \int_{S_c} dA.$$

When the classification is valid and complete,

$$a_C^{\text{flat}} + a_C^{\text{single}} + a_C^{\text{elliptic}} + a_C^{\text{saddle}} = 1.$$

These fractions answer a complementary question to I_D : not only how intense is double curvature, but how much of the surface belongs to each qualitative curvature class.

Figure 3.1 illustrates these distinctions using the analytical and semi-analytical controls from the companion manuscript. The plane and cylinder separate zero Gaussian curvature from zero mean curvature; the sphere and saddle distinguish the two signs of double curvature. The torus and half-Wigley surface introduce mixed-sign fields on closed and open geometries, respectively [1, Fig. 3 and Sec. 5.1].

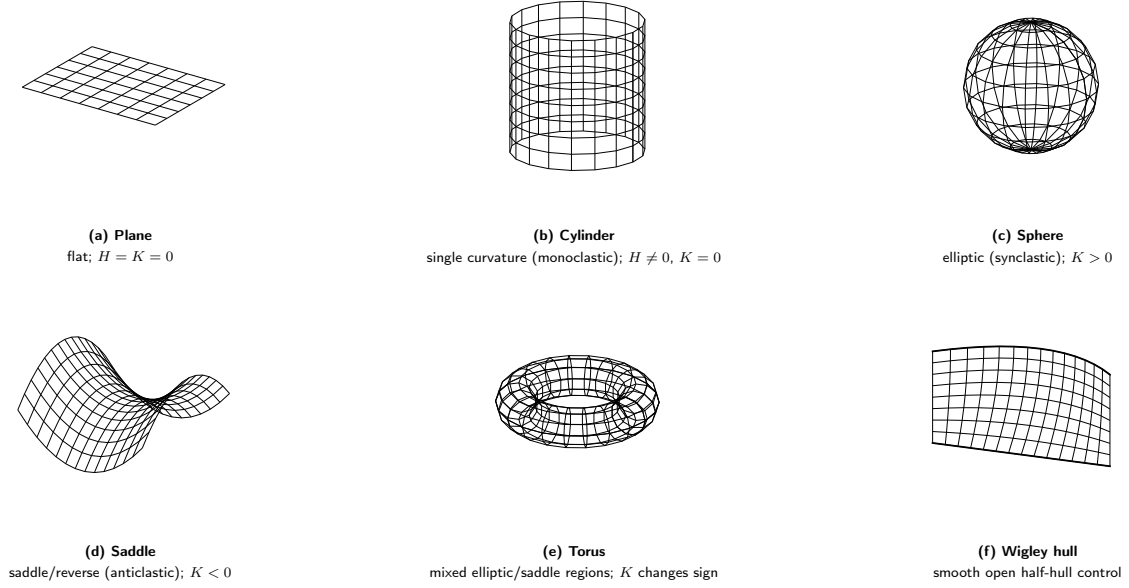


Figure 3.1.: Analytical and semi-analytical controls: plane, cylinder, sphere, saddle, torus, and half-Wigley hull. Original Figure 3 from the companion paper, p. 10 [1]; vector artwork and labels are retained. These are verification geometries, not manufacturing-cost categories.

3.5. Recommended signature

The complete v1 signature is

$$\mathbf{I}_{\text{sig}} = [I_D, I_D^+, I_D^-, (\mathbf{a}_C)^T]^T, \quad \mathbf{a}_C = [a_C^{\text{flat}}, a_C^{\text{single}}, a_C^{\text{elliptic}}, a_C^{\text{saddle}}]^T. \quad (3.3)$$

It deliberately retains three kinds of information:

- **intensity:** I_D ;
- **sign:** I_D^+ and I_D^- ;
- **areal extent:** the four components of $\mathbf{a}_C$.

The local maps of q_D , q_D^+ , q_D^- and curvature class add spatial localization to the global vector.

The seven entries have only five independent degrees of freedom: $I_D = I_D^+ + I_D^-$ and $\sum_c a_C^c = 1$. The dependent entries are retained for interpretability. A statistical model or dimensionality-reduction workflow should account for these dependencies rather than treat the seven entries as independent features [1, Sec. 3.3].

3.6. Interpretation and invariance

If a geometry is uniformly scaled by a factor s and the reference length is scaled consistently, then $K \mapsto K/s^2$ and $L_{\text{ref}}^2 \mapsto s^2 L_{\text{ref}}^2$. Therefore q_D , I_D , $I_D^\pm$ and the class thresholds expressed through the dimensionless factors remain scale consistent. Rigid translations and rotations do not change the intrinsic curvature quantities; the automatic reference-length procedure is also designed to be rigid-motion invariant.

The signature is nevertheless not independent of all modeling choices. It depends on the represented geometry, valid-domain definition, quasi-zero thresholds, and for meshes, the

differential reconstruction supported by the specific triangulation. These dependencies are therefore recorded in provenance.

3.7. Auxiliary thresholded area ratio

HullProd retains

$$I_{D,\tau} = \frac{1}{A_K} \int_{S_K} \mathbf{1}(|K| > \tau_K) \, dA$$

as `developability_area_ratio`. Under the shared classification threshold and an identical valid domain ($S_C = S_K$),

$$I_{D,\tau} = a_C^{\text{elliptic}} + a_C^{\text{saddle}}.$$

It is therefore useful as a compatibility or quick diagnostic quantity but not an independent component of the recommended signature.

Auxiliary / diagnostic

`developability_area_ratio` is intentionally excluded from $\mathbf{I}_{\text{sig}}$ because it duplicates information already present in the curvature-class fractions.

3.8. Screened experimental metrics

Earlier development considered curvature energy, curvature-gradient fairness, section waviness, FFT-based section waviness, robust variants, and local plate twist. In the v1 release these paths have distinct status:

- `curvature_energy` and `curvature_fairness`: screened experimental quantities, not computed by default;
- `section_waviness` and `section_waviness_fft`: screened experimental section diagnostics, not computed by default;
- `local_plate_twist`: mesh-dependent edge-dihedral diagnostic, not a BRep-invariant signature component;
- robust variants: research diagnostics, not replacements for the recommended v1 contract.

They can be requested with the explicit `--experimental` path, but are placed in separately labelled output sections and never enter `recommended_signature`.

3.9. Why there is no universal scalar Producibility Index

A scalar index would require either physically calibrated production/process coefficients or explicit decision weights. Those quantities are not inferable from external geometry alone. HullProd therefore preserves the vector components and their local fields rather than introducing an arbitrary universal ranking. The released output is the descriptor vector and its corresponding fields; no weighted production score is computed.

4. Software architecture and numerical implementation

4.1. Package structure

The public package keeps the user-facing pipeline compact while separating representation-specific numerical work from common reporting. The principal modules in the release are summarized below.

Module	Responsibility
<code>hullprod.api</code>	Stable <code>assess(...)</code> facade, compatibility <code>assess_hull(...)</code> , common orchestration and output writing.
<code>hullprod.backends</code>	File-extension dispatch to native BRep or triangle-mesh backends.
<code>hullprod.brep_geometry</code>	Native CAD loading, topology and direct surface differential evaluation.
<code>hullprod.brep_quadrature</code>	Deterministic adaptive quadrature over trimmed BRep faces.
<code>hullprod.brep_metrics</code>	BRep-level metric assembly, validity, topology metadata and optional display tessellation.
<code>hullprod.curvature</code>	Production mesh-curvature facade using the validated Rusinkiewicz tensor-recovery implementation.
<code>hullprod.metrics</code>	Mesh valid-domain construction, retained signature, curvature classes and optional experimental diagnostics.
<code>hullprod.mesh_quality</code>	Mesh-quality and curvature-reliability diagnostics.
<code>hullprod.reference_length</code>	Explicit/automatic L_{ref} resolution and plausibility diagnostics.
<code>hullprod.validity</code>	Common metric-validity vocabulary and human-facing labels.
<code>hullprod.export</code>	VTP/CSV distributed-field export and field manifests.
<code>hullprod.plotting</code>	Default and experimental static diagnostic plots.
<code>hullprod.report</code>	JSON/CSV/text/HTML output contract and manifests.
<code>hullprod.types</code>	<code>ProducibilityConfig</code> and <code>MetricResult</code> data contracts.
<code>hullprod.cli</code>	User-facing command-line interface and compatibility alias.

4.2. Common assessment lifecycle

The stable Python and CLI paths share the same scientific pipeline:

1. resolve the input file and output-directory policy;
2. select the backend from the file extension;
3. load the represented geometry and establish representation provenance;
4. resolve L_{ref} and curvature thresholds;
5. compute the retained metrics with backend-specific numerical methods;
6. construct metric-specific validity records;
7. generate a visualization surface and distributed fields;
8. optionally generate the two default PNG maps;
9. write signature, validity, provenance, report and manifests.

4.3. Native BRep realization

For IGES/STEP input, HullProd loads the model with the OpenCascade-based backend. The retained global metrics are computed directly on trimmed parametric faces. At a quadrature point (u, v) inside the trimmed face, the implementation evaluates surface derivatives, the local Jacobian $J(u, v)$, mean curvature $H(u, v)$ and Gaussian curvature $K(u, v)$.

For a generic density g , the surface integral on a face is evaluated in parameter space as

$$\int_{S_f} g \, dA = \int_{\Omega_f} g(u, v) J(u, v) \, du \, dv,$$

where Ω_f is the trimmed parameter domain. HullProd uses deterministic tensor-product Gauss–Legendre quadrature inside adaptively subdivided parameter cells. A lower-order and configured-order rule are compared to estimate relative change. Cells are further subdivided when the active integral quantities exceed the configured tolerance or when the quadrature cell intersects the trimmed-domain boundary.

The default BRep settings in `ProducibilityConfig` are quadrature order 5, relative tolerance 10^{-4} and maximum adaptive depth 5. Reaching the depth safeguard is not silently interpreted as convergence: the affected cell counts and integrals remain in the quadrature provenance and can trigger a cautionary or noncanonical validity status.

Canonical v1 behavior

For native CAD, the display mesh is not the source of the canonical global signature. The tessellation is generated for visualization and field transport only; BRep integrals come from direct surface derivatives and trimmed-domain quadrature.

The BRep path also records face topology, free/non-manifold edges and declared continuity across shared edges. Patch boundaries are not automatically interpreted as physical creases.

4.4. Triangle-mesh realization

For STL/OBJ/PLY input, HullProd first loads and cleans the triangulated representation. The production mesh-curvature backend uses a double-precision implementation of the Rusinkiewicz curvature-tensor recovery procedure. In summary, the algorithm:

1. constructs area-aware vertex normals;
2. fits a symmetric curvature tensor from normal variation on each triangle;
3. transports and averages the face tensors to vertices using restricted Voronoi corner areas;
4. diagonalizes the vertex tensor to recover k_1 and k_2 ;
5. forms signed $H = (k_1 + k_2)/2$ and intrinsic $K = k_1 k_2$.

The retained integrals use vertex-associated areas. Mesh boundary vertices are excluded from Gaussian-curvature validity rather than being zero-filled. Consequently,

$$A_K = \sum_{i \in \mathcal{V}_K} A_i, \quad I_D = \frac{\sum_{i \in \mathcal{V}_K} A_i |K_i| L_{\text{ref}}^2}{A_K},$$

where $\mathcal{V}_K$ is the set of valid Gaussian-curvature vertices and A_i is the associated vertex area.

Because differential recovery depends on the supplied triangulation, derivative-based mesh results are assigned the machine-readable status `mesh_representation_sensitive`. This means that the value is valid for the represented mesh, but representation-independent interpretation requires refinement or equivalent evidence.

4.5. Reference-length algorithm

The automatic L_{ref} implementation is intentionally deterministic and representation aware. For mesh input it operates on the native mesh vertices. For BRep input it operates on a controlled reference tessellation used only for geometric-span provenance. The procedure is:

1. center the sample point cloud;
2. compute its 3×3 covariance matrix;
3. sort eigenvalues in descending order;
4. if the relative leading eigengap exceeds 10^{-5} , project samples on the dominant eigenvector and take the span;
5. otherwise use twice the maximum centroid-radial distance.

The resulting metadata store the method, point count, principal variances, eigengap, and whether the isotropic fallback was used.

4.6. Metric-specific validity layer

Validity is not a single global Boolean. Each result receives a record containing a status, reason, represented area, valid area, backend, representation and method-specific convergence metadata. Important v1 statuses are:

Human-facing console and HTML reports render these states compactly, but the complete evidence remains in `validity.json` and `provenance.json`.

Table 4.2.: Principal machine-readable validity statuses.

Status	Meaning
<code>valid</code>	The value satisfies the applicable numerical and regularity contract.
<code>valid_improper_integral_convergent</code>	A singular/degenerate set exists, but nested evidence establishes a finite convergent improper integral.
<code>quadrature_unconverged</code>	Configured refinement did not establish a stable BRep integral.
<code>geometric_singularity_nonintegrable</code>	The continuous metric diverges for the represented geometry.
<code>mesh_representation_sensitive</code>	Valid for the supplied triangulation; refinement evidence is required for representation-independent interpretation.
<code>not_evaluated</code>	Optional or experimental quantity was not requested.
<code>not_applicable</code>	Quantity does not apply to the selected representation/backend.

4.7. Outputs and data contract

The `MetricResult` object exposes four main views:

- **signature:** the exact seven-component recommended v1 signature;
- **validity:** metric-specific validity records;
- **provenance:** the full representation and numerical metadata;
- **output_paths:** files generated by the shared assessment pipeline.

New integrations should begin with `signature.json`. The older raw-key `metrics.json` remains for compatibility.

Distributed fields are exported as VTK PolyData and CSV. The VTP includes developability density, positive and negative components, curvature class, validity and available H/K samples. For BRep input, those fields are carried on a display tessellation while the canonical global integral remains BRep-based.

4.8. Dependencies and release policy

The v1.0.1 runtime requires Python 3.10–3.12 and the following principal packages: NumPy, SciPy, trimesh, Matplotlib, pandas and `cadquery-ocp`. One standard installation therefore includes both mesh and native OpenCascade support. Development extras add pytest, build, Twine, Ruff and VTK.

The release follows semantic versioning: patch releases fix defects without breaking the scientific/public contract; minor releases add backward-compatible features; major releases may change public schemas, formulas, CLI behavior, field meanings or output layout.

4.9. Theory-to-code traceability

Scientific concept	Public/raw representation	Primary implementation
Total double-curvature intensity	<code>I_D</code> / <code>developability_deviation</code>	<code>metrics.py</code> ; <code>brep_quadrature.py</code>
Elliptic intensity	<code>I_D_plus</code> / positive deviation	same
Saddle/reverse intensity	<code>I_D_minus</code> / negative deviation	same
Curvature composition	<code>a_C.*</code>	<code>metrics.py</code> ; BRep class areas
Reference length	<code>reference_length</code> provenance	<code>reference_length.py</code>
Mesh curvature	H, K, principal curvatures	<code>curvature.py</code> and validated Rusinkiewicz backend
BRep curvature	direct OpenCascade differential evaluation	<code>brep_geometry.py</code> ; <code>brep_metrics.py</code>
Trimmed integration	quadrature provenance	<code>brep_quadrature.py</code>
Validity semantics	<code>validity.json</code>	<code>validity.py</code> ; BRep validity classifier
Field export	<code>surface_fields.vtp/</code> <code>csv</code>	<code>export.py</code>
User report	<code>report.html</code>	<code>report.py</code>

5. User guide

5.1. Installation

HullProd 1.0.1 supports Python 3.10–3.12. A standard installation includes both native CAD and triangle-mesh support:

```
python -m pip install hullprod
hullprod --version
```

No separate CAD extra or system OpenCascade installation is required by the documented public installation path.

5.2. First assessment

The input geometry is the root positional argument. HullProd selects the backend from the extension.

Native CAD:

```
hullprod vessel.iges
hullprod vessel.step
```

Triangulated surface:

```
hullprod vessel.stl
hullprod vessel.obj
hullprod vessel.ply
```

The default result directory is `<input-stem>_hullprod/`. The historical spelling

```
hullprod assess vessel.iges
```

remains a compatibility alias.

5.3. Choose an output directory

Use `--out` when a deterministic project location is preferred:

```
hullprod vessel.iges --out results/vessel
```

A pre-existing nonempty output directory is protected. Reuse it only deliberately:

```
hullprod vessel.iges --out results/vessel --overwrite
```

The nonempty-directory guard prevents accidental overwriting. For a separate assessment, prefer a fresh output directory rather than reusing one with `--overwrite`.

5.4. Set the reference length

Use an explicit value when a project-controlled normalization length is known:

```
hullprod vessel.iges --lref 200000
```

The number is interpreted in the geometry working unit. If the CAD file is in millimetres, 200000 means 200000 mm. HullProd prints the source unit, working unit and interpreted reference length before integration.

If no value is supplied, the automatic principal-span convention described in section 5.4.1 is used and recorded in provenance.

5.4.1. Automatic reference-length behavior

The automatic scale is a geometry-based convenience length. It is useful for zero-configuration screening and for scale-consistent descriptors, but it is not a semantic ship-design quantity such as L_{pp} . If a study requires a specific naval-architecture normalization, supply it explicitly and keep that convention fixed across all compared geometries.

5.5. Read the result directory

A normal run creates:

```
vessel_hullprod/
|-- signature.json
|-- signature.csv
|-- validity.json
|-- provenance.json
|-- summary.txt
|-- report.html
|-- metrics.json
|-- field_manifest.json
|-- output_manifest.json
|-- geometry/display_mesh.stl
|-- plots/developability_density.png
|-- plots/curvature_classes.png
'-- fields/
    |-- surface_fields.vtp
    '-- surface_fields.csv
```

A practical reading order is:

1. open `report.html` for the compact human-readable overview;
2. inspect `signature.json` or `signature.csv` for the retained numbers;
3. verify `validity.json` before treating a finite value as canonical;
4. retain `provenance.json` with every published or optimization-ready signature;
5. use `surface_fields.vtp` or the PNG maps to identify where the global values originate.

Important limitation

Do not copy only the seven numbers into a paper or optimization database and discard the validity/provenance record. The scientific meaning of a mesh signature, the selected L_{ref} , thresholds, valid area and numerical status are part of the result.

5.6. Interpret the recommended signature

The public JSON and Python property use the semantic names:

```
I_D
I_D_plus
I_D_minus
a_C.flat
a_C.single
a_C.elliptic
a_C.saddle
```

For a valid classification, the four `a_C` fractions should sum to one up to numerical round-off. Also check the identity $I_D \approx I_D^+ + I_D^-$.

The most useful interpretation sequence is:

1. compare I_D for total double-curvature intensity;
2. inspect I_D^+ versus I_D^- to identify elliptic versus saddle/reverse contribution;
3. compare the four class fractions to understand areal composition;
4. inspect the local developability and class maps to locate the responsible regions;
5. check validity and representation provenance before drawing cross-geometry conclusions.

5.7. Visualize distributed fields in ParaView

Open `fields/surface_fields.vtp` in ParaView, click **Apply**, and choose a point-data array from the coloring menu. Useful arrays include developability density, its positive/negative components, curvature class and available H/K values. The accompanying `field_manifest.json` documents each exported array and its interpretation.

For BRep input, remember that the VTP surface is a display/transport tessellation. It is suitable for localization and visualization, but it is not the source of the canonical global BRep integral.

5.8. Python API

The stable public facade uses the same backend dispatch and output writers as the CLI:

```
1 from hullprod import assess
2
3 result = assess("myvessel.iges")
4 print(result.signature)
5 print(result.validity)
6 print(result.provenance)
```

Write the complete artifact tree:

```

1 from hullprod import assess
2
3 result = assess(
4     "myvessel.iges",
5     out_dir="myvessel_hullprod",
6     plots=True,
7 )

```

Specify a reference length and intentionally disable PNG generation while retaining machine-readable fields and reports:

```

1 result = assess(
2     "myvessel.iges",
3     lref=142.0,
4     out_dir="results",
5     plots=False,
6 )

```

`assess_hull(...)` and `ProducibilityConfig` remain available for advanced and compatibility use.

5.9. Worked example: assessing the bundled sphere

This example uses `simple_sphere.iges`, an analytical geometry distributed with HullProd v1.0.1 under `hullprod/examples/`. The package configuration includes the IGES and STL fixtures in the installed distribution. The example therefore requires neither a private acceptance-test report nor an external hull-geometry download. It creates a new assessment in a user-selected working directory.

5.9.1. Prepare a supported Python environment

Use CPython 3.10, 3.11 or 3.12. For example, from a working directory on macOS or Linux with Python 3.12 installed:

```

python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install "hullprod==1.0.1"
python -m hullprod.cli --version

```

On Windows, use PowerShell with Python 3.12 installed:

```

py -3.12 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install "hullprod==1.0.1"
python -m hullprod.cli --version

```

A Git clone is not required for the installed-package workflow below.

5.9.2. Run the example from the installed package

Save the complete code below as `sphere_example.py` in the chosen working directory. This is a script to create from the listing, not a file assumed to exist in the public repository. It locates the geometry through Python package resources, uses an explicit reference length, and writes the standard assessment artifacts.

```

1 import json
2 from importlib.resources import as_file, files
3 from pathlib import Path
4
5 from hullprod import __version__, assess
6
7 if __version__ != "1.0.1":
8     raise RuntimeError("This example targets HullProd 1.0.1.")
9
10 geometry = files("hullprod.examples").joinpath(
11     "simple_sphere.iges"
12 )
13 if not geometry.is_file():
14     raise FileNotFoundError("Bundled sphere geometry not found.")
15
16 out = Path("sphere_example").resolve()
17 if out.exists() and (not out.is_dir() or any(out.iterdir())):
18     raise FileExistsError("Choose a new or empty output directory.")
19
20 with as_file(geometry) as sphere_path:
21     result = assess(
22         sphere_path,
23         lref=2.0,
24         out_dir=out,
25         plots=True,
26         progress=print,
27     )
28
29 print(json.dumps(result.signature, indent=2, allow_nan=False))
30 print(json.dumps(result.validity, indent=2, allow_nan=False))
31 print("Open the report:", out / "report.html")

```

Execute the saved script with the activated environment:

```
python sphere_example.py
```

The output directory is `sphere_example/` relative to the directory from which the script is run. An existing nonempty output directory is deliberately rejected rather than silently overwritten. To repeat the assessment, select a different output directory in the script.

5.9.3. Inspect the generated artifacts

Open `sphere_example/report.html` in a browser. Read the global values from `signature.json` or `signature.csv`, and inspect `validity.json` and `provenance.json` before interpreting them. The two standard PNG maps are under `plots/`; `fields/surface_fields.vtp` and `fields/surface_fields.csv` carry the distributed fields. These files are generated by the assessment; they are not pre-existing reports that must be downloaded from the repository.

For this native-BRep example, the global integrals come from direct CAD derivatives and trimmed-domain quadrature. The display tessellation carries sampled fields for visualization, not the canonical integral. An entirely elliptic sphere should give a uniform curvature-class map; the developability-density field should be constant up to numerical error. The default density plot uses its documented logarithmic display transform, so its colorbar should not be read as the raw value of q_D .

5.9.4. Compare against analytical references

The bundled IGES sphere has radius $R = 1$ mm. With the explicit reference length $L_{\text{ref}} = 2$ mm used above,

$$K = \frac{1}{R^2}, \quad q_D = |K|L_{\text{ref}}^2 = 4, \quad I_D = I_D^+ = 4, \quad I_D^- = 0.$$

The entire classified surface is elliptic. The corresponding analytical targets are summarized in table 5.1, consistently with the unit-sphere control in the companion paper [1, Table 4].

Table 5.1.: Analytical reference values for a unit-radius native-BRep sphere with $L_{\text{ref}} = 2$ in matching length units. These are theoretical targets, not values transcribed from a new execution.

Quantity	Reference value	Interpretation
Surface area	$4\pi \simeq 12.56637061 \text{ mm}^2$	Entire sphere surface
I_D	4	Total double-curvature intensity
I_D_plus	4	Elliptic contribution
I_D_minus	0	No saddle contribution
a_C.flat	0	No flat classified area
a_C.single	0	No single-curvature classified area
a_C.elliptic	1	Entire classified area is elliptic
a_C.saddle	0	No saddle classified area

Check $I_D \simeq I_D^+ + I_D^-$ and $\sum_c a_C^c \simeq 1$, then compare the computed values with the analytical targets using appropriate numerical tolerances. Read the actual metric-specific statuses from `validity.json`; finite numbers alone are not an acceptance criterion. Display triangle counts, timing and final decimal digits are not prescribed expected output. Omitting `lref=2.0` selects the automatic reference-length procedure and changes the normalization convention.

Verification scope. This example is a self-contained procedure using a public packaged geometry and the released API. Its code has been checked against the v1.0.1 source and checked for Python syntax; no new execution in a supported Python environment is reported here. The values above are analytical references. Numerical outputs and validity must be established by running the procedure.

5.10. Experimental quantities

The default assessment keeps screened research quantities out of the scientific signature. Request them only when a research task specifically needs them:

```
hullprod vessel.iges --experimental
```

They appear in separately labelled JSON/report sections and never enter `recommended_signature`. This is intentionally different from treating every computable number as a production-ready metric.

5.11. Skip plots

The two default PNGs can be omitted without losing the VTP/CSV fields or machine-readable results:

```
hullprod vessel.iges --no-plots
```

5.12. Exit behavior

Assessment returns exit code 0 when the run completes and artifacts are written, even if an individual validity record is cautionary, such as a mesh-sensitive value. Fatal input, parser, backend, I/O or runtime failures return a nonzero exit code. Therefore automated workflows should inspect both process success and metric validity.

5.13. Common mistakes

- **Treating automatic L_{ref} as L_{pp} .** It is a geometric principal-span convention, not inferred naval-architecture metadata.
- **Comparing mesh and BRep values as if the numerical realization were identical.** They answer the same metric definition through different representations and carry distinct validity semantics.
- **Ignoring units for `--lref`.** The explicit value is in the input working unit.
- **Interpreting h_f and k_f as manufacturing limits.** They are numerical quasi-zero classification factors.
- **Publishing a finite value without its validity record.** Finite does not automatically mean canonical.
- **Using the experimental set as the v1 signature.** Curvature energy/fairness and section metrics were deliberately screened out of the recommended contract.
- **Calling the output a production-cost model.** HullProd is geometry screening only.

6. Verification and benchmark examples

6.1. Evidence used in this manual

The analytical and public-hull results are reported in the companion paper [1]. The original paper figures retain their artwork and legends. The BRep–mesh comparison chart uses an unchanged benchmark CSV archived at the public HullProd v1.0.1 tag. The separate sphere example in section 5.9 is an executable procedure using the geometry bundled with v1.0.1; its tabulated values are analytical references, not a new numerical result. Artifact paths and hashes for the archived campaign are listed in chapter D.

The frozen release changelog records clean-wheel checks on Linux with Python 3.10–3.12 and on macOS Apple Silicon with Python 3.12, covering native BRep and mesh inputs. Version 1.0.1 is a metadata-only patch over 1.0.0. These software checks do not constitute validation against shipyard production measurements.

6.2. Analytical and semi-analytical controls

The controls in fig. 3.1 separate the main geometric questions before any public-hull interpretation. The plane and cylinder have zero Gaussian curvature but different mean curvature; the sphere isolates positive Gaussian curvature; the saddle isolates negative Gaussian curvature; and the torus and half-Wigley surface exercise spatially varying, mixed-sign fields [1, Sec. 5.1].

Table 6.1.: Selected native-BRep verification values from Table 4 of the companion paper [1]. Errors are relative for the nonzero references. The saddle is a sign control, not an independently quantified scalar-accuracy reference in that table.

Control	Reference I_D	Native BRep I_D	Reported comparison
Plane	0	0	Exact zero; flat class
Unit sphere	4.00000000	4.00000000	2.22×10^{-16}
Cylinder	0	0	Exact zero; single class
Saddle	$I_D > 0$	0.746436	$I_D^+ = 0, I_D^- = I_D$
Torus	13.58122181	13.58122181	2.62×10^{-16}
Half-Wigley	4.30990339	4.30989710	1.46×10^{-6}

The torus uses major radius $R = 3$, minor radius $r = 1$ and $L_{\text{ref}} = 8$. It has equal positive and negative contributions, $I_D^+ = I_D^- = 64/(3\pi)$. The half-Wigley control has $L = 1$, $B = 0.1$, $T = 0.0625$ and $L_{\text{ref}} = L$. Its surface is defined by

$$y = \frac{B}{2} \left[1 - \left(\frac{2x}{L} \right)^2 \right] \left[1 - \left(\frac{z}{T} \right)^2 \right], \quad -L/2 \leq x \leq L/2, \quad -T \leq z \leq 0.$$

The manuscript’s independent semi-analytical reference gives $I_D^+ = I_D^- = 2.15495169$. Its native result is 2.15494850 for each contribution. These values refer to the particular control and quadrature protocol, not a universal accuracy guarantee [1, Tables 4 and Sec. 6.1].

6.3. Public-hull geometries and comparison convention

The public-hull campaign comprises DTMB 5415, KCS, JBC and KVLCC2M. Figure 6.1 reproduces the original geometry view. The campaign normalizes by the documented L_{pp} , not by the automatic principal span used by a zero-configuration CLI assessment. The half/full representation and reference length in table 6.2 must therefore accompany comparisons [1, Table 3].

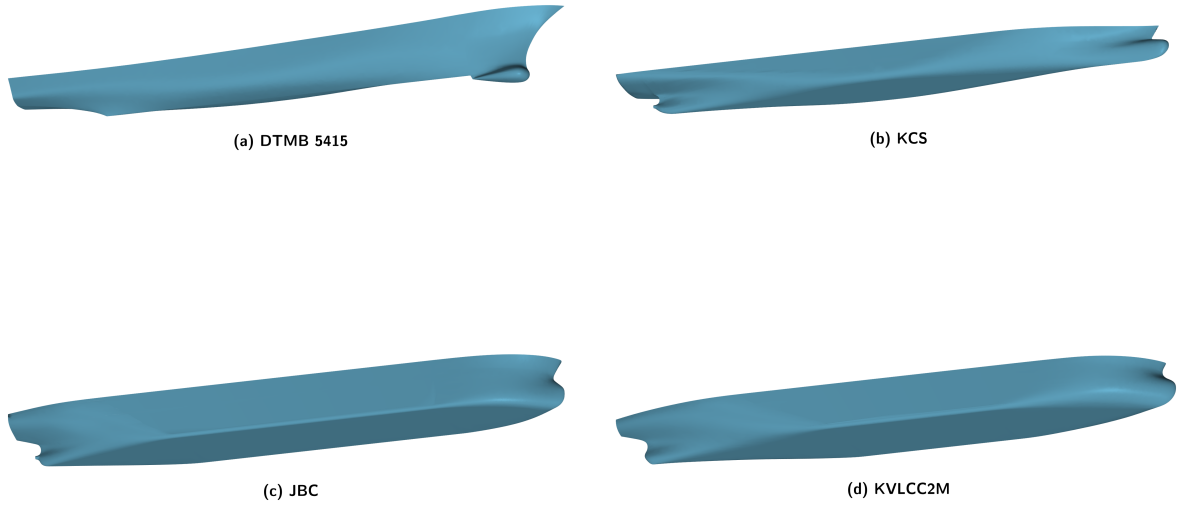


Figure 6.1.: Public-hull geometries: (a) DTMB 5415, (b) KCS, (c) JBC and (d) NMRI faired KVLCC2M. Original Figure 4 from the companion paper, p. 11 [1]. Original geometry proportions, panel labels and common orthographic view are retained; only exterior white margins have been removed.

Table 6.2.: Public-hull reference lengths and source domains from Table 3 of the companion paper [1]. Values are in the documented model scale.

Case	$L_{\text{ref}} = L_{pp}$ [m]	Authoritative represented domain
DTMB 5415	5.72	Starboard half hull
KCS	7.2786	Starboard half bare hull
JBC	7.00	Full symmetric bare hull
KVLCC2M	4.97	NMRI faired full symmetric bare hull

6.4. Global signatures: intensity and areal extent

Tables 6.3 and 6.4 reproduce the native-BRep part of the manuscript’s Table 5. The asterisk qualifies the developability integral: it denotes a convergent improper integral rather than a fully regular field everywhere. The class fractions are normalized by the valid classified area of the corresponding representation [1].

Table 6.3.: Native-BRep total and signed developability from manuscript Table 5 [1]. An asterisk denotes a convergent improper developability integral. Values are reproduced at the manuscript’s printed precision.

Hull	I_D	I_D^+	I_D^-
DTMB 5415	66.4011*	33.6190	32.7821
KCS	82.6084	41.1121	41.4963
JBC	58.6029*	22.6040	35.9989
KVLCC2M	53.0006*	31.6765	21.3241

Table 6.4.: Native-BRep represented-valid-area fractions from manuscript Table 5 [1]; $h_f = k_f = 10^{-4}$. Small departures of printed row sums from one are due to tabular rounding.

Hull	Flat	Single	Elliptic	Saddle
DTMB 5415	0.00000	0.00003	0.51225	0.48772
KCS	0.21266	0.05839	0.32758	0.40138
JBC	0.50066	0.15527	0.15515	0.18892
KVLCC2M	0.54009	0.02490	0.27006	0.16495

KCS has approximately 24% greater I_D than DTMB 5415, although its elliptic and saddle regions occupy approximately 72.9% of valid area rather than nearly the complete valid area. This is the intended distinction between intensity and areal extent. It does not establish that KCS is more expensive or more difficult to manufacture [1, Sec. 6.2].

The original campaign includes metric-specific regularity evidence. Native BRep developability is reported as convergent improper for DTMB 5415, JBC and KVLCC2M, and finite/converged for KCS. The native valid-area fraction is reported as 100% to the table’s precision, with singular-set provenance retained where applicable [1, Table 7].

6.5. Distributed developability: locating the contribution

A global average does not identify where double curvature is concentrated. Figure 6.2 retains the manuscript’s direct-BRep field samples and its common display scale. The bow, stern, bilge, keel and individual CAD patches can contribute differently even when the global intensities are comparable [1, Sec. 6.3].

These panels are manuscript visualizations, not a screenshot of the default CLI report. Their common robust color range makes the four hulls readable together. It must not be confused with either a manufacturing limit or a numerical cutoff. The triangles carry direct BRep differential samples for display only; they do not define the canonical global BRep integral. Very large isolated display samples therefore require the recorded integral/regularity evidence, not interpretation from the colorbar alone [1, Fig. 5 and Sec. 6.3].

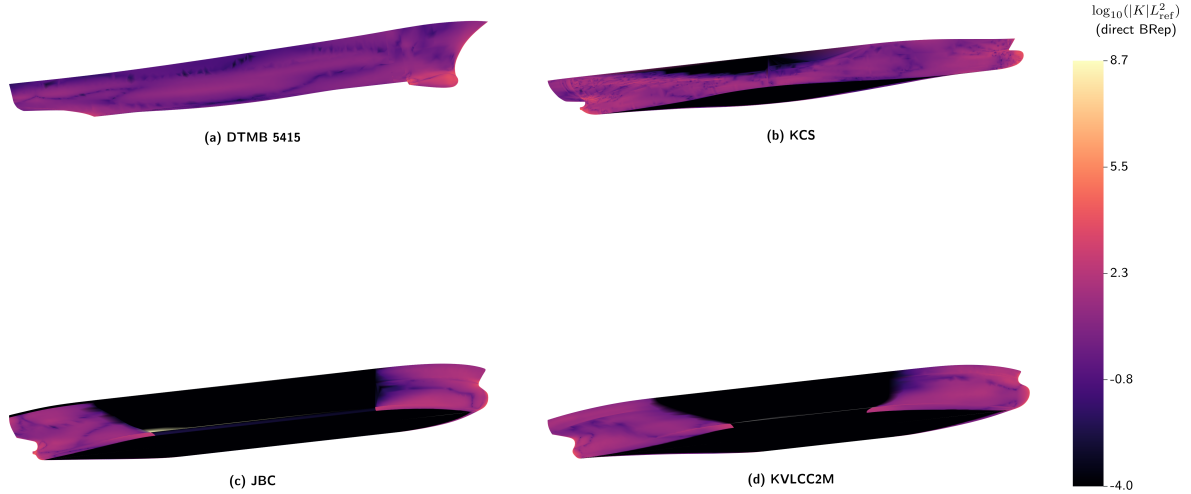


Figure 6.2.: Distributed native-BRep density $q_D = |K|L_{\text{ref}}^2$ for the four public hulls. Original Figure 5 from the companion paper, p. 14 [1]. The common display range is $-4 \leq \log_{10}(q_D) \leq 8.692$: lower and upper tails are graphically saturated only, and invalid samples are grey. The lower bound is the quasi-zero scale $k_f = 10^{-4}$; the upper bound is the pooled 99.5th percentile of positive samples. No clipping is applied to integration or reported metrics.

6.6. Distributed curvature classes: reading areal composition

Figure 6.3 provides the spatial counterpart of the four fractions in table 6.4. The original colorbar is retained: invalid grey, flat light grey, single curvature green, elliptic blue, and saddle/reverse red. These are the manuscript’s figure colors; the numerical class meanings, rather than a color choice, define the scientific result [1, Fig. 6].

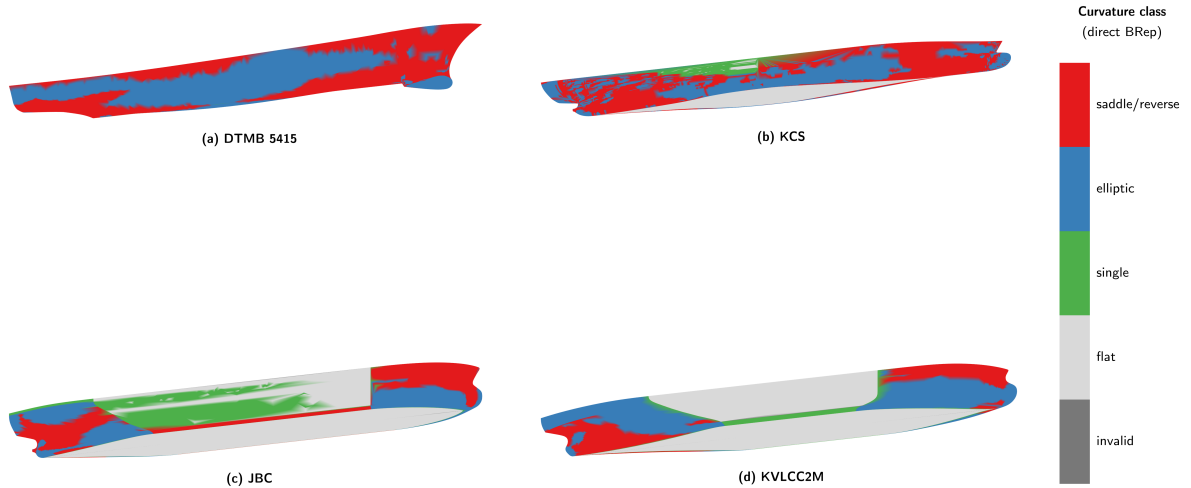


Figure 6.3.: Distributed native-BRep curvature classes for DTMB 5415, KCS, JBC and KVLCC2M. Original Figure 6 from the companion paper, p. 15 [1]. The fields distinguish invalid, flat, singly curved, elliptic and saddle/reverse regions. Class fractions aggregate represented valid classified area under the declared quasi-zero thresholds; these are geometric classes, not calibrated forming-cost categories.

The same visual region need not dominate both kinds of descriptor. A small region with intense curvature can materially contribute to I_D , whereas a large quasi-flat region contributes strongly to areal composition but little to the developability numerator. Local maps should therefore be

read alongside total/signed intensities, class fractions, valid area and representation provenance rather than used as independent production-feasibility maps [1, Sec. 6.3].

6.7. Actual archived output: native BRep versus fine STL

The public HullProd repository at tag `v1.0.1` contains the benchmark CSV with all eight native-BRep/fine-STL signature rows. Its repository-relative path and Git blob hash are recorded in chapter D. Figure 6.4 plots those unchanged archived values; it is not a new HullProd calculation.

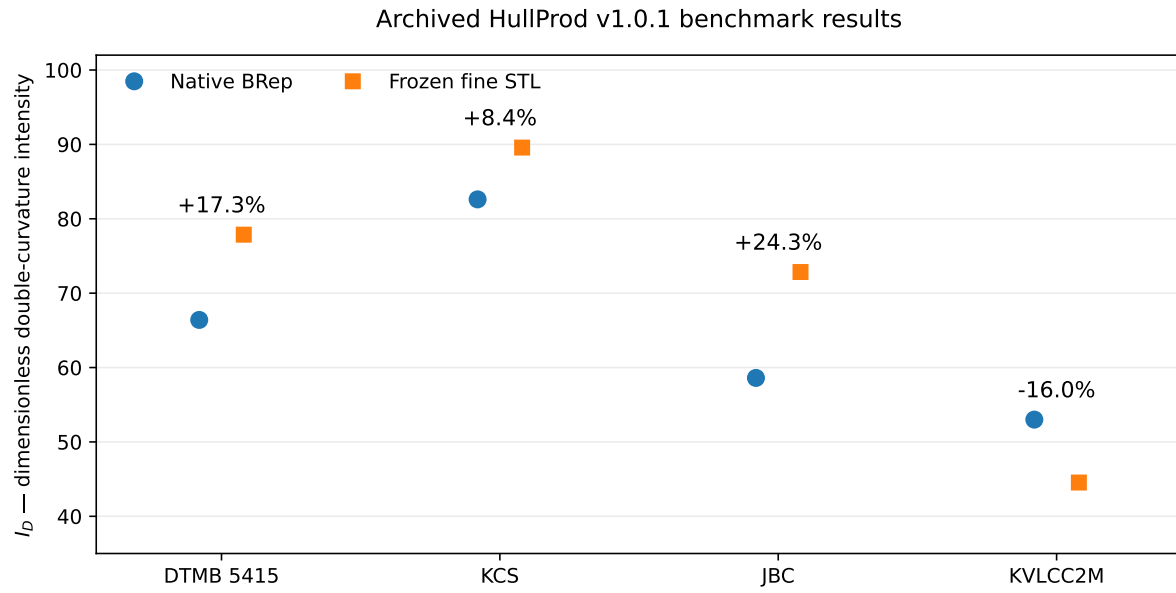


Figure 6.4.: Total developability intensity from the actual archived v1.0.1 benchmark CSV. Annotations show $100(I_D^{\text{STL}}/I_D^{\text{BRep}} - 1)$. The plot preserves the manuscript case order and is not a ranking of production difficulty. DTMB 5415, JBC and KVLCC2M native values have the convergent-improper qualification reported in manuscript Table 5 [1].

Table 6.5.: Selected fields from the archived release CSV. Values are rounded here for readability; the source CSV in the versioned public repository retains the original precision. Every native-BRep valid-area fraction in that archive is 1.0.

Hull	Native I_D	Fine STL I_D	Difference [%]	STL valid area [%]
DTMB 5415	66.4011	77.8628	+17.3	90.683
KCS	82.6084	89.5729	+8.4	93.697
JBC	58.6029	72.8435	+24.3	97.350
KVLCC2M	53.0006	44.5400	-16.0	95.365

These are comparisons of the actual campaign representations, not isolated pointwise curvature-recovery errors. They include tessellation, resolution, valid-domain, boundary and local mesh-quality effects. The separate controlled same-face study evaluates matching physical vertices and a common interior mask; its reported fine-level Gaussian-curvature errors range from approximately 1.4% to 8.2%. That result does not establish whole-hull convergence of the integrated signature [1, Tables 5–6 and Sec. 6.4].

6.8. Validity and consistency checks

Native quadrature convergence and local cell diagnostics serve different roles. A retained global result may be valid while local bounded-depth cells produce a quadrature caution. An unresolved finite-depth value, however, is not an established canonical integral. A convergent improper integral also remains distinct from a non-integrable geometric singularity. Mesh results retain the representation-sensitive status and require their mesh-quality and refinement evidence [1].

Before reusing a signature, check input identity, backend, reference-length convention and unit, metric-specific status, valid area, thresholds, and quadrature or mesh provenance. Then verify $I_D \simeq I_D^+ + I_D^-$, $\sum_c a_C^e \simeq 1$ on the established classified domain, and whether local maps reveal unexpected invalid or concentrated regions. Finite values alone are insufficient.

For reproducibility, retain the original geometry or immutable permitted source, software version/tag, nondefault settings, `signature.json`, `validity.json`, `provenance.json`, and the VTP/CSV fields used in distributed interpretation. This manual reproduces paper figures and archived numerical values; the external providers' hull-geometry files are not included.

6.9. What remains outside the verification claim

The evidence supports the retained geometry-only descriptors and their representation-aware interpretation for the reported controls and hulls. It does not calibrate fabrication cost, man-hours, forming route, plate-scale feasibility or shipyard schedule. Nor does it validate a universal scalarization of the signature. Those require additional production/process information and independent physical evidence, not missing values inferred from external geometry [1, Secs. 7–8].

7. Scope and reporting practice

7.1. Meaning of the assessment

HullProd characterizes the represented external hull surface using curvature intensity, sign and areal composition. The seven entries of the recommended vector remain separate; the package supplies no universal aggregate score. The numerical quasi-zero factors define classification thresholds at the stated reference scale, not calibrated production-acceptance criteria [1].

Identify the assessed surface explicitly: bare or appended, half or complete hull. Additional surfaces change the curvature distribution and area normalization. HullProd evaluates the supplied representation rather than inferring missing geometry.

7.2. Keep the numerical qualifications with the result

The global signature is read together with `validity.json` and `provenance.json`. Preserve the input hash, backend, reference-length value and unit, classification thresholds, represented and valid areas, and numerical settings. A finite value labelled `mesh_representation_sensitive` is an assessment of that triangulation, not evidence of a representation-independent limit. A `quadrature_unconverged` record is not replaced by the last finite sample.

The signed-intensity identity $I_D = I_D^+ + I_D^-$ and the class identity $\sum_c a_C^c = 1$ apply on their respective valid domains. These identities are consistency checks, not convergence evidence. The vector consequently has five independent degrees of freedom despite its seven reported entries [1, Sec. 3.3].

7.3. Use local maps with their scales

Read the developability map as the localization of $q_D = |K|L_{\text{ref}}^2$ and the categorical map as the distribution of the four geometric classes. Retain each figure's legend, treatment of invalid samples and display range. In the paper examples, graphical saturation is a visualization choice only; the numerical integrals are not clipped. A BRep display tessellation carries differential samples but does not define the canonical global integral.

7.4. Boundary of the software claim

The present assessment does not predict fabrication cost, labor hours, forming route, production schedule or shipyard-specific feasibility. It does not perform hull optimization, panelization, seam placement or forming simulation. A lower value of a geometric descriptor is therefore not a general ranking of manufacturing difficulty [1].

A. Public API and CLI map

A.1. Public Python symbols

The top-level package exports the principal v1 user-facing objects:

Symbol	Role
<code>assess</code>	Stable shared assessment facade for native BRep and mesh inputs.
<code>assess_hull</code>	Advanced compatibility assessment entry point.
<code>MetricResult</code>	Container exposing signature, validity, provenance and output paths.
<code>ProducibilityConfig</code>	Advanced numerical/configuration contract.
<code>analyze_mesh_quality</code>	Mesh-quality diagnostic utility.
<code>SCHEMA_VERSION</code> , <code>FIELD_SCHEMA_VERSION</code> , <code>OUTPUT_LAYOUT_VERSION</code>	Versioned machine-readable contracts.
<code>__version__</code>	Package version.

A.2. Primary CLI

The visible user-facing command is

```
hullprod INPUT [--out DIR] [--lref VALUE] [--overwrite]
              [--no-plots] [--experimental] [--quiet] [--debug]
```

Option	Meaning
<code>INPUT</code>	IGES/STEP BRep or STL/OBJ/PLY triangle surface.
<code>--out DIR</code>	Output directory; defaults to <code>./<input-stem>_hullprod</code> .
<code>--lref VALUE</code>	Authoritative reference length in the geometry working unit.
<code>--overwrite</code>	Permit writing into a nonempty output directory.
<code>--no-plots</code>	Omit the two default PNG maps; machine-readable fields remain.
<code>--experimental</code>	Compute the separately labelled screened research set.
<code>--quiet</code>	Suppress progress messages on stderr.
<code>--debug</code>	Show a traceback for unexpected failures.
<code>--version</code>	Print the installed HullProd version.

Several additional expert/development controls exist in `ProducibilityConfig` and the CLI implementation for thresholds, BRep quadrature, display tessellation, sections and robustness studies. They are intentionally hidden from normal help because changing them changes the numerical experiment. Use them only when the study design explicitly requires and records those choices.

B. Output and schema map

The result-directory tree is shown in the user guide. This appendix provides a compact reference to each file's role; the stable signature must always be interpreted together with validity and provenance.

B.1. File roles

File	Intended role
<code>signature.json</code>	Stable structured v1 result including the recommended signature and hierarchy of supporting information. New software integrations should start here.
<code>signature.csv</code>	One-row table for pandas, spreadsheets, surrogate-model inputs and optimization pipelines.
<code>validity.json</code>	Metric-specific scientific/numerical validity and reasons.
<code>provenance.json</code>	Input identity, versions, backend/representation, reference length, thresholds, area coverage, numerical settings, quality/topology, units, warnings and timings.
<code>summary.txt</code>	Concise plain-text run/signature summary.
<code>report.html</code>	Self-contained user report; not a substitute for raw validity/provenance records.
<code>metrics.json</code>	Raw/compatibility key view retained for established users.
<code>field_manifest.json</code>	Meaning, association, formula and validity of exported surface arrays.
<code>output_manifest.json</code>	Inventory of files produced by the run.
<code>display_mesh.stl</code>	Surface carrying output fields; visualization transport for BRep, assessed representation for mesh input.
<code>developability_density.png</code>	Static map of $q_D = K L_{\text{ref}}^2$ with invalid regions distinguished.
<code>curvature_classes.png</code>	Stable categorical map of flat, single, elliptic, saddle/reverse and invalid regions.
<code>surface_fields.vtp</code>	ParaView-readable VTK PolyData containing distributed fields.
<code>surface_fields.csv</code>	Portable coordinates and practical field columns for generic tools.

C. Software snapshot and traceability

C.1. Documented software version

Item	Recorded value
Repository	<code>cnr-inm-mao/hullprod</code>
Software tag	<code>v1.0.1</code>
Commit	<code>0015ed350aefc004f649811d4c3a823bb23777ea</code>
Release date	3 September 2026
Package name	<code>hullprod</code>
Supported Python	CPython 3.10–3.12
Software license	BSD-3-Clause
Manual edition	1.0.1-r6; 24 September 2026

Version 1.0.0 established the documented scientific and software behavior. Version 1.0.1 corrects the funding acknowledgement without changing the numerical definitions, API, CLI or output schemas. This edition documents the frozen software tag rather than assigning current `main` changes to that release.

C.2. Implementation traceability

The release README, changelog, `pyproject.toml`, public API and versioned documentation identify the executable conventions. The principal implementation paths are listed in Chapter 4. They identify where the equations and output meanings are realized; they are not additional scholarly references.

C.3. Scientific reference and evidence identity

The companion paper by Serani and Maki [1] is the single scientific reference for the metric formulation and benchmark interpretation. The release commit, input hashes and artifact paths identify the software and data used in the examples. The sphere tutorial uses the public v1.0.1 package fixture and explicit normalization; the user generates the corresponding output directory by running the complete procedure in section 5.9.

The public benchmark CSV location and its Git blob hash are reported in chapter D, together with the origins of the figures and the scope of the public sphere tutorial.

D. Figure and example provenance

D.1. Figure inventory

The paper figures are reproduced from the manuscript version supplied for this manual. Their source numbering and page locations are recorded below; all scientific citations refer to the same paper [1].

Table D.1.: Figure provenance and editorial treatment.

Manual figure	Origin	Treatment
Figure 3.1	Paper Fig. 3, p. 10	Vector crop of original analytical-control artwork.
Figure 6.1	Paper Fig. 4, p. 11	Original image; exterior white margins removed.
Figure 6.2	Paper Fig. 5, p. 14	Original image, labels and color scale retained; exterior margins removed.
Figure 6.3	Paper Fig. 6, p. 15	Original categorical image and legend retained; exterior margins removed.
Figure 6.4	Release benchmark CSV	Plot of unchanged archived numerical values; no new HullProd assessment.

Archived release data. The BRep-mesh comparison uses the unchanged CSV archived in the public HullProd repository at tag v1.0.1. Its repository-relative path is:

```
benchmarks/results/retained_signature_closing_audit/brep_fine_retained_
signature_comparison.csv
```

The versioned file is available from the [public repository](#). Its Git blob SHA-1 is:

```
246e2e560e5cc83fac1533e516f31180421f5e19
```

Figure 6.4 plots these archived numerical values without reporting a new HullProd assessment.

Public sphere tutorial. The procedure in section 5.9 uses `hullprod/examples/simple_sphere.iges` from the public v1.0.1 package. The complete script is printed in the section and creates the assessment artifacts through the released API. The tabulated values are analytical targets; the procedure has not been executed for this manual edition. No private report is needed to run it.

Reference

- [1] A. Serani and K. J. Maki (2026). *Geometry-Based Metrics for Early-Stage Hull-Form Producibility Screening*. arXiv preprint arXiv:2609.27544 [cs.CE]. <https://arxiv.org/abs/2609.27544>. DOI: [10.48550/arXiv.2609.27544](https://doi.org/10.48550/arXiv.2609.27544).

Software and document identification

This manual edition is 1.0.1-r6, describing HullProd v1.0.1 at commit 0015ed35. The manual revision changes documentation only; it does not change the package, metric definitions or numerical results. Numerical-data sources and illustration provenance are recorded in the appendices.

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